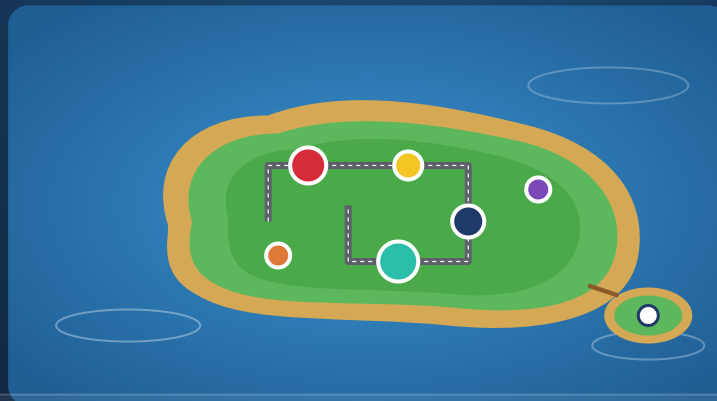


LEGO ISLAND

PROCESS BOOK • MILESTONE 3



Project Team — COM-480 Data Visualization

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OVERVIEW & MOTIVATION

LEGO has released over 18,000 sets across five decades, spanning 154 distinct themes ranging from classic Town and Castle to Star Wars, Harry Potter, and the rise of premium adult collector sets. However, most existing analyses treat this rich cultural legacy with boring analytical charts and reports.

We asked: What if exploring this history felt like actually *playing* with the bricks themselves? Our answer is **LEGO Island**: a fully interactive island map inspired by classic exploration games, where every geographic district represents a major lego theme. Users wander around the island by clicking districts to reveal animated data panels, and explore through subthemes down to the individual lego sets.

CORE DESIGN PRINCIPLE

The physical characteristics are metaphor, not decoration. Everything *gives meaning*. District logos get bigger/smaller with set count in the selected time window, geographic proximity hints at thematic relationships, and the ocean separates the main island from the smaller island dedicated to Licensed lego sets. Geography becomes the primary manifestation of data, charts become secondary support.

THE DATASET AT A GLANCE

18,457 total records

12,757 "Normal" sets used

154 total themes

52 years (1970–2022)

8 island main themes

WHO IS THIS FOR?

The target audience is intentionally broad: collectors, parents researching sets, anyone interested in lego-data. The interface is super accessible and allows casual exploration as much as focused analysis. A first-time visitor can wander the island purely visually, while a data enthusiast can use the year slider, drill through subtheme data, and read the four supporting charts and featured sets in detail.

WHAT YOU'LL FIND IN THIS BOOK

02

DATASET & PREPROCESSING

Sourcing, filtering, the missing-data story, and why we chose median alongside mean.

03

SKETCHES & EVOLUTION

Milestone 1 hand sketches, expanded with annotations on what changed and why.

04

CHALLENGES

The four biggest problems we hit — zoom origin maths, parallax conflicts, sparse price data.

05

ARCHITECTURE & TEAM

How the seven JavaScript modules fit together, and who built what.

DATASET & PREPROCESSING

We used the LEGO Sets dataset from Maven Analytics, compiled from the community-maintained Brickset encyclopedia. It contains 14 fields per record across every official set released between 1970 and 2022.

FIELDS

set_id name year theme subtheme themeGroup category pieces minifigs ageRange_min
US_retailPrice bricksetURL thumbnailURL imageURL

THE FOUR BIG DATA DECISIONS

01

MISSINGNESS PATTERN

`US_retailPrice` is missing for 59.5% of sets, almost entirely pre-2000 sets where historical prices were never catalogued. `ageRange_min` missing for 54.6%, `minifigs` for 40.5%. We reflect this honestly in the UI rather than imputing.

02

FILTERING STRATEGY

We filter to `category = "Normal"`, excluding Lego Merch&Gear, Books, Promotional items that are not normal lego sets. This brings our working set to 12,757 records with substantially better data quality.

03

SKEW & CENTRAL TENDENCY

Mean piece count is 238 but median is only 79: strong right skew from collector megaset (max piece count: 11,695). We show median *and* mean in every stats row to avoid misleading users about a "typical" set.

04

IMAGE COVERAGE

95% of Normal sets include an `imageURL`. This high coverage promoted the "featured sets card gallery" from "nice-to-have" to a must-have feature, giving the website its rich visual presentation with hundreds of real images.

WHAT WE FOUND IN THE EDA

Annual releases grew from roughly 65 a year in the 1970s to over 500 a year by 2015-21. The 2010s alone account for over 7,400 sets, more than every previous decade combined. Mean piece count grew from ~99 pieces in the 1980s to ~472 in the 2020s, showing the rise of the adult interest to more complex sets. Among the 5,164 sets with both price and pieces, retail price correlates strongly with pieces ($r = 0.867$), and the most expensive sets are Star Wars Ultimate Collectors Series models.

THE PIPELINE

All processing happens client-side at runtime. We fetch the CSV, then call a single `aggregate()` function that aggregates every set into theme → subtheme → year lookup tables, and computes per-district statistics (set count, mean/median piece counts, min/max year, yearly retail price, etc.), and stores a featured-set list sorted by piece count descending. The result is cached on a global `store` object, so panel opens are near-instant. No further computation over the 12,757 rows are needed once the page has loaded.

MILESTONE 2 SKETCHES, EXPANDED

Our Milestone 2 submission proposed three core sketches. We reproduce and annotate them below, marking what survived, what changed, and what was added.



Fig. 1 – M2 sketch: top-down island overview. District size = set count. Click → drill-down panel.

Changed: added logos, added roads, trees, rocks, small island... Added DATA STORY website, yearly slider, removed tooltip
Kept: top-down view, general island structure, ocean compass & theme organization

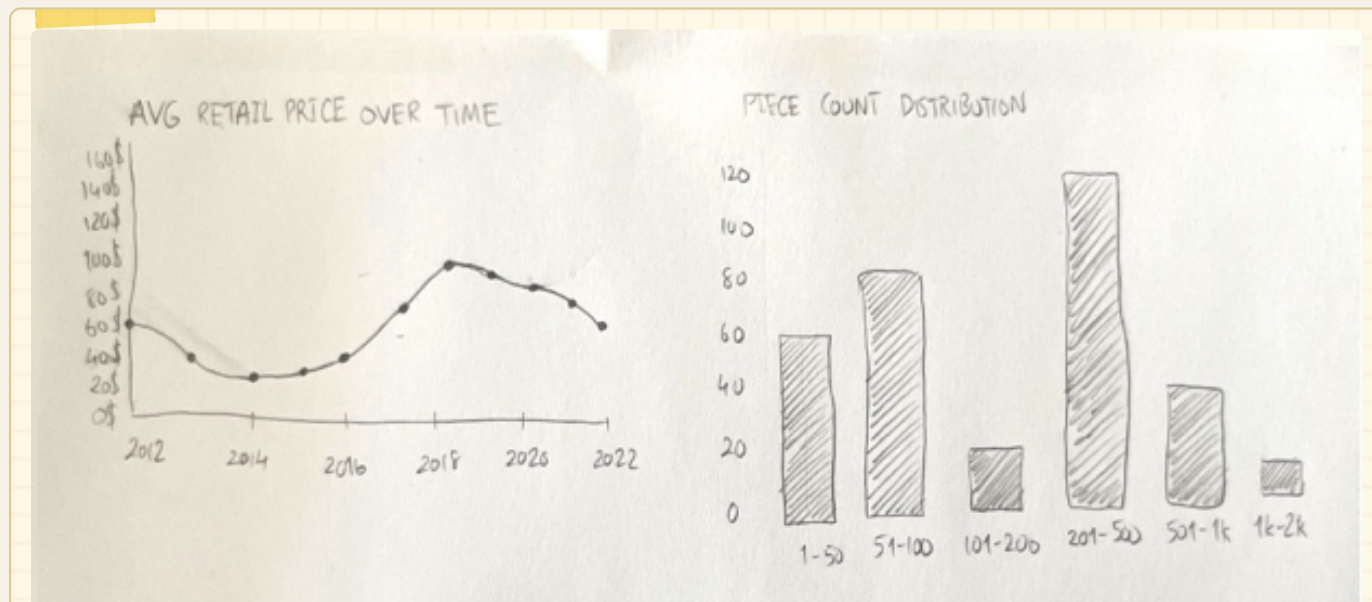


Fig. 2 – M2 sketch: hand-drawn chart studies. Left: avg retail price over time. Right: piece-count histogram. Both shipped, both got polish in D3.

Kept: both chart types exactly as drawn. Line for price, bars for piece count.
Added in code: sets released per year line chart, and first row of key statistics (total sets, years active, avg pieces, price range)

THE PATH WE **ACTUALLY** TOOK

Five major pivots stand between the Milestone 2 sketches and what was produced. Every change aimed for cleaner data presentation and simpler interaction while having more places for users to look, more reasons to keep exploring.

M1 • MARCH

Island concept established

Top-down island, with clickable district theme bubbles to open a data panel. The single core goal that survived throughout the project: *use geography as the primary data visualization*, not just charts.

M2 • APRIL

Panel pivot: side-split → bottom sheet

The Mileston2 MVP involved a split-view layout where data panel came out from the right side and covered half of the page. A data panel placed at the bottom presents the data in a more spacious manner, works on narrower screens, and creates a more visually beautiful "reveal" moment as the island zooms toward the clicked district behind it.

MID-DEV • APRIL

SVG → Custom Background Image

Our early plan was to generate the island as an SVG with drawn terrain. We switched to a custom background image and layered the interactive district badges on top as absolutely positioned DOM elements. This separation of concerns static art beneath, live data layer above was cleaner to develop and easier to tune visually.

MID-DEV • MAY

The "IP folder" pattern

Licensed sets (Star Wars, Harry Potter, Marvel...) deserved their own district but contain too many sub-franchises to show. Therefore, we invented a folder district: clicking it opens a card grid of franchises, and clicking a franchise loads its full data. Two-level navigation kept the panel from overflowing.

FINAL • MAY

Year range slider, Districts sized proportionally by set count

Originally a static view, we added a yearly range slider that filters the entire island data in real time. Every district marker rescales in size with sqrt-area proportional scaling based on the number of sets published in that timeframe. This single addition transformed the island from a map into a *time travel explorer*, a great way to tell a story with our data.

FINAL • MAY

Per-district visual identity

Late in development we introduced a per-district `accentColor` variable. The chart colors are different in every district's data panel. The interface stops feeling like one design and starts feeling like 8 different designs for each subtheme. In addition, we adjusted final district positions and the road network to reflect thematic similarities (Creator and Technic are close together, Licensed theme has its own small island since it's an endless world on its own, etc.) between districts.

WHAT WE DROPPED

The Milestone 2 plan listed more extra plans we didn't implement. Two of those: comparison mode and the guided-tour 2D character were cut. Comparison mode duplicated a lot of the work we already did and wouldn't look good visually. The tour mode wouldn't contribute to the storytelling since users prefer clicking directly, so we couldn't justify it against simpler wins like the year slider.

CHALLENGES & DESIGN DECISIONS

Every creative design generates challenges along the way. Below are the four problems that cost us the most time, and the resolutions we settled on. Each one shaped the final product in a meaningful way.

CHALLENGE - YEAR SLIDER BUBBLE SIZE

District set counts vary by up to 20× — Duplo has over 1,200 sets while Architecture has under 100. Using a linear naïve CSS `transform: scale()` made small districts nearly invisible; a fixed size ignored the data entirely.

▼ SOLUTION - SQRT BUBBLE SCALING

We applied square-root scaling to the set count ratio, then mapped it to a badge scale range of 0.6–1.5, doing `scale(calc(var(--badge-scale, 1) * 1.12))`. Square root gives area-proportional size: a district with 4× the sets appears roughly 2× as wide, which feels natural without the largest themes completely dominating the island.

CHALLENGE - ANIMATION DIFFICULTY

The island needed many layered animations running simultaneously: water ripples, marker entrance stagger, parallax on mouse move, zooming to district, panel slide-up, chart tooltips... All without breaking or conflicting with each other.

▼ SOLUTION - ANIMATION LAYERING

Each animation system is isolated: water runs on its own canvas RAF loop; parallax uses a separate RAF with lerp smoothing; CSS animation delay handles marker entrance and chart stagger declaratively. Parallax is explicitly paused before the zoom transition fires, then resumed 520ms later once the CSS transition has settled preventing the two systems from fighting over the island's transform.

CHALLENGE - BUBBLES IN WRONG POSITIONS

District badges are positioned using percentage coordinates derived from the SVG viewBox. On different screen sizes and aspect ratios, district bubbles moved off their intended positions on the island background image, breaking the spatial placement

▼ SOLUTION - RESPONSIVE POSITIONING

We used absolute positioning on top of the custom background image, placing each bubble at a left/top percentage that matched its intended landmark. The key here was ensuring the background image always scaled at a fixed aspect ratio so the percentage positions stayed locked to the right spots regardless of screen size, rather than moving as the image stretched or shrunk unevenly.

CHALLENGE - FINDING VISUAL ASSETS

The project needed a large collection of visual assets: a custom island background that looked hand-crafted and on-brand, plus logos and imagery for all 8 districts and 6 IP sub-franchises — none of which existed as ready-made resources.

▼ SOLUTION - ASSET COLLECTION

The island background was generated using an AI image tool and we iterated until it matched the LEGO Island game aesthetic with roads, trees, bridges, and enough visual variation to make the island look unique and playful. District badge photos were sourced from official LEGO set image library. Franchise logos (Star Wars, Ninjago, Harry Potter, Marvel, Chima, Bionicle) were downloaded individually from brand websites and converted to consistent formats for use in the website.

ARCHITECTURE & EXTRAS

DELIVERED

We chose vanilla JavaScript. The project loads directly in a browser over a local HTTP server.

MODULE	RESPONSIBILITY	KEY FUNCTIONS
<code>data.js</code>	CSV loading (PapaParse), aggregation of all theme and subtheme data at startup	<code>initData()</code> , <code>store</code>
<code>island.js</code>	District marker DOM generation, zoom-to-district transform, dynamic marker scaling	<code>initIsland()</code> , <code>updateMarkerSizes()</code>
<code>panel.js</code>	Bottom-sheet open/close, normal vs Licensed panel, statistics row HTML, featured sets card grid layout	<code>openPanel()</code> , <code>closePanel()</code>
<code>charts.js</code>	All four D3 charts: sets released per year chart, piece count histogram, price trend chart, subtheme bar charts. Info tooltips.	<code>renderCharts()</code> , <code>clearCharts()</code>
<code>parallax.js</code>	Mouse move RAF parallax on island and district layers	<code>pauseParallax()</code>
<code>water.js</code>	Canvas ripple animation, randomised spawn zones, radius growth, fade animation. Runs before data load.	<code>initWater()</code>
<code>main.js</code>	Bootstrap sequence: water → data → parallax → island → year slider → hide loading screen	<code>initYearSlider()</code>

CSS IN 7 SMALL FILES

`variables.css` `reset.css` `layout.css` `island.css` `panel.css` `charts.css` `animations.css`

Loaded in dependency order. Tokens live only in `variables.css`; nothing else hardcodes a colour. Animations are isolated so they can be disabled via `prefers-reduced-motion`.

EXTRAS: WHAT SHIPPED VS. WHAT WAS DROPPED

EXTRA 1

YEAR RANGE SLIDER

Dual-thumb filter; sqrt-area marker rescaling; live "# of sets in window" counter. **SHIPPED**

EXTRA 6

ANIMATED WATER

Canvas ripples spawn at 10 randomised ocean zones. Runs *before* data arrives. **SHIPPED**

NEW

LICENSED FOLDER PATTERN

Licensed sets grouped under a folder district. Click → franchise grid → charts. **SHIPPED**

MVP+

SET CARD GALLERY

Up to 24 featured sets per district, sorted by piece count desc., lazy-loaded, with error fallback.

SHIPPED

NEW

PER DISTRICT COLOR THEME

Each theme has a primary & accent colour for their charts and featured sets card. **SHIPPED**

E4

COMPARISON MODE

Side-by-side theme comparison charts. **NOT DONE**

THE DATA STORY SITE

Alongside the island, we built a secondary website "Experience the LEGO Story" that delivers the dataset as **five chapters of one narrative**. Where the island invites free exploration, this website walks the visitor through a fixed presentation: from **when** themes started, to **what shape** the LEGO universe has, to **which themes** grew in time.

THE FIVE CHAPTERS

CH 1

HEATMAP

Theme group × 5-year period. Colour = set count. The licensed explosion of the 2000s is impossible to miss.

CH 2

NIGHTINGALE ROSE

One petal per theme group. Petal length (sqrt scale) shows how much of the catalogue each one really owns.

CH 3

TREEMAP

Drill-down rectangles: theme group → individual subthemes. Click any group to zoom in; breadcrumb to go back.

CH 4

CHORD DIAGRAM

Theme groups + piece-size tiers. Surprising finding: licensed sets tend to be *smaller*, not bigger.

CH 5

SUNBURST CHART

Two-ring radial: groups (inner) → island themes (outer). Click inner = zoom in; click outer = open set gallery.

EXTRA

SETS MODAL

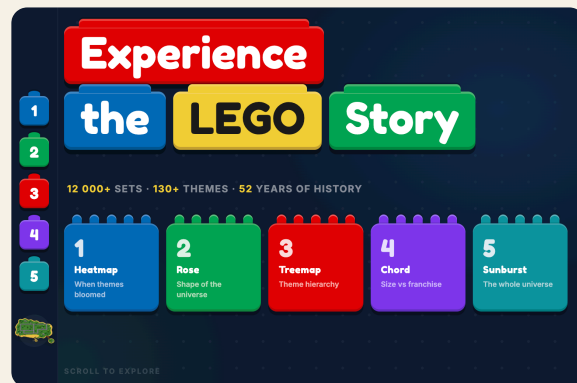
Click an individual theme anywhere and a bottom sheet slides up with up to 24 featured sets, same component the island uses.

AGE & PROCESS

The hero title is built from **physical brick-word tiles**. Each word is its own coloured brick. The dark navy backdrop reuses the island's lego pattern so the two surfaces feel like one continuous world.

A **fixed left sidebar** turns the five chapters into brick-shaped quicklinks with floating labels. They slide right on scrolling so the visitor always knows where they are and when clicked, move the user to that section. It also has a special island button to go back to the island

Every chart card carries a coloured top accent strip matching the chapter's brick colour. The same one-style-drives-everything pattern we used on the island.



↑ Initial View of `story.html`: lego-brick title, chapter-brick tiles, side nav. of chapter shortcuts.

TWO WAYS TO EXPLORE DATA

The island is for wandering around. Its job is to make the visitors curious. On the other hand, the data story is for reading. Its job is to make visitors understand. They present the same data, but in a different way. Island prioritizes exploration and playfulness, website prioritizes storytelling and analytical analysis.

TEAM & PEER ASSESSMENT

Work was divided across three areas: island rendering & interaction, data pipeline & charts, and panel UI & CSS. All three members contributed to design decisions from Milestone 1 onward and checked each other's code. The split below shows primary ownership, everybody worked on all features together and contributed equally.

K

Daniel Sura

ISLAND & INTERACTION

- ▶ Island custom image & district placement
- ▶ District bubble DOM
- ▶ Zoom-to-district parallax logic
- ▶ Did the "DATA STORY" website
- ▶ Year slider UI & district rescaling
- ▶ CSS variables & layout architecture

A

Altu Bozyaka

D3 & CHARTS & PROCESS BOOK

- ▶ D3 chart suite all four chart types
- ▶ Shared tooltip system
- ▶ Licensed folder & franchise cards
- ▶ Stats row HTML & formatting
- ▶ Chart CSS
- ▶ Wrote the Process Book
- ▶ Asset collection

D

Kian Hampson Bahia

DATA ANALYSIS & PIPELINE

- ▶ CSV loading & PapaParse integration, data cleaning&analysis
- ▶ Aggregation pipeline (`aggregate()`)
- ▶ Piece histogram bins & price trend
- ▶ Featured set card grid & image fallback
- ▶ Water ripple and jetski animation, JS animations, loading page
- ▶ Did the "DATA STORY" website

CONTRIBUTION BALANCE

Effort was roughly equal across the three members (33% each). The functional split is by primary ownership; the overall design vision was developed collectively from the Milestone 1 brainstorm and the code was written together in common study sessions.

LEGO ISLAND • COM-480 • 2026

BUILT BRICK BY BRICK.

KIAN • ALTU • DANIEL

